

Identify the lesion before treatment

Pain, blur and a corneal infiltrate with epithelial loss suggest microbial keratitis requiring urgent assessment.



The photograph shows dense central corneal opacity and surrounding haze in a red eye. In a symptomatic patient, check fluorescein for epithelial loss and assess the anterior chamber; appearance alone cannot establish the organism.

Take the risk history

Ask about contact lenses, overnight wear, cleaning and water exposure, as well as injury, previous surgery and steroid use. Contact lenses particularly raise concern for *Pseudomonas*. Obtain corneal scraping for Gram stain and culture to guide treatment. A quiet, well-defined old scar is different from an acutely painful infiltrate with surrounding haze. Intense anterior chamber inflammation can produce a hypopyon, a layer of inflammatory cells that heightens concern about disease severity.

Treat intensively and judge the response

Microbial ulcers can deteriorate quickly; frequent therapy and close reassessment are central to care.

Treatment sequence

Urgent ophthalmic assessment and corneal microbiological sampling



Intensive broad-spectrum fortified topical antibiotic therapy



Modify antimicrobial choice using culture and sensitivity



Review healing, thinning and intraocular complications closely

When to increase support

Consider admission for severe infection, disease in the only functioning eye or inability to administer frequent drops. Stop contact lens wear and address predisposing surface disease. Systemic antibiotics may be needed when sclera is involved or severe thinning threatens perforation, under specialist care. Ulceration can progress to perforation and endophthalmitis; healed lesions may leave scarring and irregular astigmatism that reduce vision.